```
   return queue, rDists
                                                    #13
```

#1 Method `updateQueue` takes a point `p`, a kdTree with all the points in the dataset, the queue of points reachable from current cluster, an array with the points that have already been processed, an array with the dataset's reachability distances, and the parameters epsilon and min points. It returns a tuple with the queue and the array of reachability distances (possibly updated during this call).

#2 Perform the range query collecting all points in the hyper-sphere with center `q`.

#3 Check if `p` is a core point (if and only if the query at line #2 returns at least `minPts` points). If it isn't, return without performing any action.

#4-#5 Cycle through all unprocessed points in `p`'s ε-neighborhood.

#6 Compute the reachability distance of `q` from `p`.

#7 Check if the reachability distance of `q` from the previously processed points is `null`.

#8-#9 If it is, `q` hasn't been added to the queue yet, so insert it now, and set the reachability distance for `q`.

#10 Otherwise, check if the reachability distance of `q` from `p` is smaller than from the previously processed points (meaning, if `p` is closer to `q` than those points).

#11-#12 If that's the case, update `q`'s reachability distance and its priority in the queue.

#13 Return the updated queue and array of reachability distances.